Input Shaping Control of an Overhead Crane with Time-varying Cable Length using a Generalized Input Shaper

Ho Duc Tho[1], Kazuhiko Terashima[2], and Takanori Miyoshi[1]

***Abstract*— In this brief, a new input shaping-based control law is proposed for an overhead crane with time-varying cable length. The proposed approach can accommodate a generalized input shaper and general hoisting motions. It will be shown that a non-robust input shaper can be used within the established scheme to completely suppress residual vibration at the completion of a maneuver, outperforming the standard (robust) input shaping controllers. Simulation results are provided.**

I. Introduction

Hoist, which is an essential motion in overhead cranes, poses a significant challenge for the classical input shaping control. The reason is that the hoisting motion introduces time-varying cable length into the crane dynamics, making a complete vibration suppression unachievable when using classical input shaping controllers. This fact was reported in [1]. Until now, the main method to mitigate the adverse effect caused by time-varying dynamics is to use robust input shaper types such as zero derivatives reinforcement [2], [3], extra insensitivity (EI) [3]–[6], and optimized specified insensitivity (SI) [7], [8]. Recently, an attempt on employing an artificial neural network trained by a particle swarm optimization mechanism has been made in [9]. This approach aims at modifying the unity-magnitude zero-vibration input shaper online to alleviate the unpredictability of the rope length and the payload mass. Nevertheless, none of the existing studies using (arbitrarily robust) input shaping technique is able to totally eliminate vibration for overhead crane systems with general time-varying rope length, namely that, only a fraction of vibration can be reduced. This is the main problem to be resolved in this paper.

In this paper, an input shaping-based control law will be put forward to enable a complete vibration suppression for overhead crane systems with time-varying cable length caused by the hoisting motion. A general time-polynomial will be employed to describe the hoisting motion. Based on the Weierstrass approximation theorem [10, p. 148], a general polynomial with a sufficiently high order can approximate any continuous function to an arbitrary degree of precision, and thus, a general hoisting motion can be attained. Moreover, a generalized input shaper with an adjustable order of derivative constraints can also be integrated into the proposed input shaping-based controller to enhance robustness of the transportation if needed. In the case that motion commands are accurately executed by the crane's actuators, a non-robust input shaper can even be used to significantly reduce the transfer time while guaranteeing the perfect zero-vibration condition at the completion of the maneuver.

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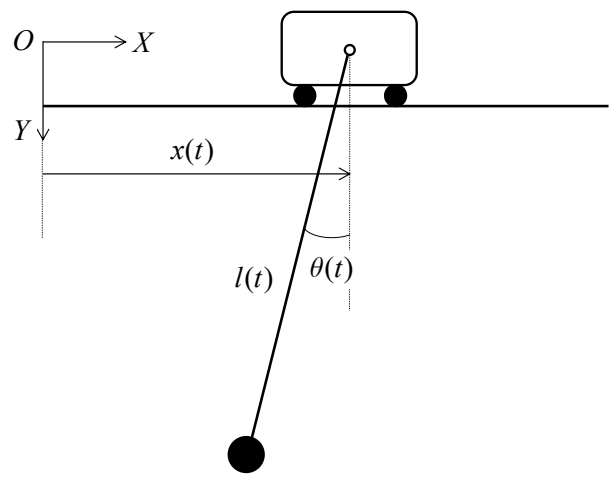

Fig. 1. An overhead crane with time-varying cable length.

To clarify, this paper is a further development of [11] and [12] on the following points:

- By using a nth-order time-polynomial, general hoisting motions can be realized. Therefore, the transportation becomes more flexible.
- A generalized input shaper is employed, thus, arbitrary robustness with respect to parametric uncertainties can be achieved.
- The solvability of the proposed technique is rigorously discussed (see Remark 1).

II. Problem formulation and design concept

A. Problem formulation

A mathematical model of an overhead crane with hoisting motion is illustrated in Fig. 1. The cart position, the sway angle, and the time-varying rope length are denoted as $x(t)$ (m), $\theta(t)$ (rad), and $l(t)$ (m), respectively. The system parameters include M (the trolley mass, in kg), m (the payload mass, in kg), and g (the gravitational acceleration, $g = 9.81$ m/s^2). The kinetic energy $\mathcal{K}$, potential energy $\mathcal{V}$, and Lagrangian function $\mathcal{T}$ of the system are

$$\mathcal{K} = \frac{1}{2}M\dot{x}^2 + \frac{1}{2}m\left\{\left[\frac{d}{dt}\left(x + l\sin\theta\right)\right]^2 + \left[\frac{d}{dt}\left(l\cos\theta\right)\right]^2\right\}$$

$$\mathcal{V} = -mgl\cos\theta$$

$$\mathcal{T} = \mathcal{K} - \mathcal{V}.$$

The Euler-Lagrange equation applying to the unactuated coordinate $\theta(t)$ reads

$$\frac{d}{dt}\left(\frac{\partial \mathcal{T}}{\partial \dot{\theta}}\right) - \frac{\partial \mathcal{T}}{\partial \theta} = 0. \tag{1}$$

By developing (1), dynamics (in the kinematic level) of the overhead crane is governed by

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} + g\sin\theta = -\cos\theta\ddot{x}. \tag{2}$$

The following assumptions are made throughout the paper

- The payload starts from rest [13]–[15], that is to say

$$x(0) = \dot{x}(0) = \dot{l}(0) = \theta(0) = \dot{\theta}(0) = 0. \tag{3}$$

- The sway angle is sufficiently small so that the nonlinear system (2) can be approximated by the following linear time-varying system [16]–[18]

$$l\ddot{\theta} + 2\dot{l}\dot{\theta} + g\theta = u, \tag{4}$$

  where $u = -\ddot{x}$, i.e., the acceleration $-\ddot{x}$ is the control input of the system.

Under the above assumptions, we wish to design an input shaping-based control law $u(t)$ such that for some finite *shaped* terminal time $\tau_f > 0$, the following terminal conditions, which are based on practical requirements of the transportation, should be guaranteed

1) At τ_f, the cart reaches a desired destination at zero speed, namely that

$$x(\tau_f) = x_d,\ \ \dot{x}(\tau_f) = 0, \tag{5}$$

   where x_d is a constant reference for the cart.
2) The cable length $l(t)$ should pass through nodes at predefined time instants and also stops at τ_f, i.e.,

$$l(t_i) = l_i,\ \ \dot{l}(t_{n-2}) = 0,\ i = 0, \ldots, n-2, \tag{6}$$

   where $n \in \mathbb{N}$ ($n \geq 3$, $\mathbb{N}$ is the set of integers), and t_i and l_i are given in advance with a requirement that $t_i \neq t_j$ for all $i \neq j$ ($i, j = 0, \ldots, n-2$). Moreover, let $t_0 = 0$ and $t_{n-2} = \tau_f$.
3) At τ_f, the vibration ceases, namely that

$$\theta(\tau_f) = 0,\ \ \dot{\theta}(\tau_f) = 0. \tag{7}$$

In the next subsection, a control design concept will be introduced to realize the above-mentioned control requirements.

B. Design concept

The basic design concept of the proposed approach is shown in Fig. 2. As mentioned in section I, the input shaper should not be directly applied to the time-varying system (4) since this will result in a poor vibration suppression performance. Alternatively, the standard input shaping controller is firstly applied to a reference model, which is an undamped second-order linear time-invariant (LTI) system. This step produces a reference oscillation that is exactly zero at the end of the maneuver. The actual control input of the system (4) is then devised to perfectly match the vibration of the real system with that of the reference oscillation. To

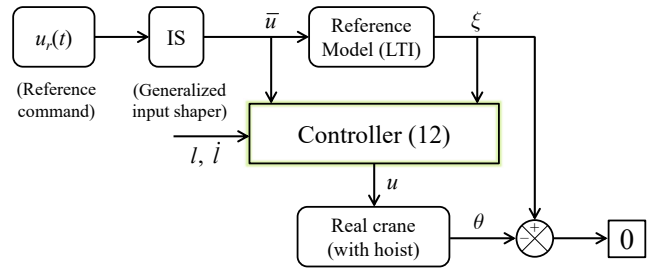

Fig. 2. The design concept of the proposed input shaping-based controller.

accomplish the design task, the reference unshaped command $u_r(t)$ (see Fig. 2) must be determined to fulfill control objectives other than the vibration suppression. This can be done by parameterizing $u_r(t)$ in an appropriate manner. In the subsequent section, such a design concept will be elaborated.

III. Input shaping-based controller design

We firstly choose the following undamped reference model

$$L\ddot{\xi} + g\xi = \bar{u}(t), \tag{8}$$

where $L > 0$ is a constant nominal rope length, $\xi(t)$ is the reference oscillation, and $\bar{u}(t)$ is the input of the reference model. The natural angular frequency of the reference model is $\omega = \sqrt{g/L}$ and its natural period is $T = 2\pi/\omega$. The standard input shaping control law is applied to (8), i.e.,

$$\bar{u}(t) = u_r(t) \star I(t), \tag{9}$$

where $\star$ stands for the convolution operator. Denote s as the complex variable, C_m^k as a k-combination from m elements, and $\delta(.)$ as the Dirac delta distribution, the generalized input shaper $I(t)$ in (9) is given by [19]

$$I(t) = \frac{1}{2^m}\sum_{k=0}^{m} C_m^k \delta\left(t - \frac{kT}{2}\right), \tag{10}$$

whose Laplace transform is $\mathcal{I}(s) = \left(1 + e^{-sT/2}\right)^m / 2^m$. Note that, $m - 1 \in \mathbb{N}$ is the order of zero-derivative constraints of the generalized input shaper. A higher value of m means that the input shaper is more robust to uncertainties. The unshaped command $u_r(t)$ is parameterized as follows

$$u_r(t) = \begin{cases} \beta_1 t + \beta_0, & t \in [0,\ t_f] \\ 0, & t > t_f, \end{cases} \tag{11}$$

where t_f—the *unshaped* terminal time—is a positive real number chosen by the designer, and β_i ($i = 0, 1$) are to-be-determined variables to realize the control objective (5). When the control input (9) is applied for the reference model (8), the reference oscillation $\xi(t)$ is suppressed at the shaped terminal time $\tau_f = t_f + mT/2$, i.e., $\xi(\tau_f) = \dot{\xi}(\tau_f) = 0$. The next step to design the real control input $u(t)$ to match the actual vibration $\theta(t)$ with the reference oscillation $\xi(t)$. To achieve this goal, the following control law is proposed

$$u(t) = g\left(1 - \frac{l}{L}\right)\theta + 2\dot{l}\dot{\theta} + \frac{l}{L}\bar{u}. \tag{12}$$

$$
\begin{aligned}
P(t) = \; & \sum_{k=0}^{m} C_m^k H\left(t - \frac{kT}{2}\right) - \sum_{k=0}^{m} C_m^k H\left(t - t_f - \frac{kT}{2}\right) + \sum_{k=0}^{m} (-1)^k C_m^k \cos(\omega t) H\left(t - \frac{kT}{2}\right) - \\
& \sum_{k=0}^{m} (-1)^k C_m^k \cos(\omega(t - t_f)) H\left(t - t_f - \frac{kT}{2}\right)
\end{aligned} \tag{17}
$$

$$
\begin{aligned}
Q(t) = \; & \sum_{k=0}^{m} C_m^k \left(t - \frac{kT}{2}\right) H\left(t - \frac{kT}{2}\right) - \sum_{k=0}^{m} C_m^k \left(t - t_f - \frac{kT}{2}\right) H\left(t - t_f - \frac{kT}{2}\right) - \\
& t_f \sum_{k=0}^{m} C_m^k H\left(t - t_f - \frac{kT}{2}\right) + \frac{1}{\omega} \sum_{k=0}^{m} (-1)^k C_m^k \sin(\omega t) H\left(t - \frac{kT}{2}\right) - \\
& \frac{1}{\omega} \sum_{k=0}^{m} (-1)^k C_m^k \sin(\omega(t - t_f)) H\left(t - t_f - \frac{kT}{2}\right) - t_f \sum_{k=0}^{m} (-1)^k C_m^k \cos(\omega(t - t_f)) H\left(t - t_f - \frac{kT}{2}\right).
\end{aligned} \tag{18}
$$

Substituting (12) into (4), the following result can be obtained

$$
L\ddot{\theta} + g\theta = \bar{u}. \tag{13}
$$

Comparing the above equation with (8), if we choose $\xi(0) = \theta(0)$ and $\dot{\xi}(0) = \dot{\theta}(0)$, a perfect model matching can be attained. Therefore, by using the control law (12), the real oscillation is completely suppressed at the terminal time τ_f, that is to say, $\theta(\tau_f) = \dot{\theta}(\tau_f) = 0$. By this, the control objective (7) is fulfilled.

Next, the hoisting motion will be designed to realize the control objective (6). The cable length function $l(t)$ is chosen to be a nth-order time-polynomial as follows

$$
l(t) = \alpha_0 + \alpha_1 t + \ldots + \alpha_n t^n = \sum_{i=0}^{n} \alpha_i t^i. \tag{14}
$$

Projecting the control objective (6) onto (14), the coefficients α_i $(i = 0, \ldots, n)$ can be determined by solving the following system of linear equations

$$
\begin{bmatrix}
0 & 1 & 2t_0 & 3t_0^2 & \ldots & nt_0^{n-1} \\
0 & 1 & 2t_{n-2} & 3t_{n-2}^2 & \ldots & nt_{n-2}^{n-1} \\
1 & t_0 & t_0^2 & t_0^3 & \ldots & t_0^n \\
1 & t_1 & t_1^2 & t_1^3 & \ldots & t_1^n \\
\vdots & & & & & \\
1 & t_{n-2} & t_{n-2}^2 & t_{n-2}^3 & \ldots & t_{n-2}^n
\end{bmatrix}
\begin{bmatrix} \alpha_0 \\ \alpha_1 \\ \alpha_2 \\ \alpha_3 \\ \vdots \\ \alpha_n \end{bmatrix}
=
\begin{bmatrix} 0 \\ 0 \\ l_0 \\ l_1 \\ \vdots \\ l_{n-2} \end{bmatrix} \tag{15}
$$

The solution of (15) is closely related to the generalized Hermite interpolation problem, and it was shown in [20] that such a solution always exists and is unique. The remaining control objective (5), which is associated to the cart motion, will be discussed in the next presentation.

Since the control input $u(t)$ is designed in the acceleration level, we have to perform integrations on (4) to determine the cart's position and speed. In order to do so, a closed-form solution of the real oscillation $\theta(t)$ needs to be known in advance. For this subtask, Laplace transform can be implemented. Denote $\mathcal{L}$ as the Laplace transform operator, the Laplace transform $\mathcal{U}_r(s)$ of the unshaped command $u_r(t)$ can be computed as follows

$$
\begin{aligned}
\mathcal{U}_r(s) = \mathcal{L}\{u_r(t)\} &= \int_0^{t_f} (\beta_0 + \beta_1 t) e^{-st} dt \\
\Longrightarrow \mathcal{U}_r(s) &= \frac{\beta_0}{s} + \frac{\beta_1}{s^2} - \left(\frac{\beta_0 + \beta_1 t_f}{s} + \frac{\beta_1}{s^2}\right) e^{-st_f}.
\end{aligned}
$$

Applying the Laplace transform operator on both sides of the linear integro-differential equation (13) results in

$$
\Theta(s) = \mathcal{P}(s)\beta_0 + \mathcal{Q}(s)\beta_1,
$$

where $\Theta(s)$ is the Laplace transform of $\theta(t)$, and $\mathcal{P}(s)$ and $\mathcal{Q}(s)$ are given by

$$
\begin{aligned}
\mathcal{P}(s) &= \frac{(1 - e^{-st_f})\left(1 + e^{-sT/2}\right)^m}{2^m s\,(Ls^2 + g)} \\
\mathcal{Q}(s) &= \frac{\left[1 - (1 + t_f s)\, e^{-st_f}\right]\left(1 + e^{-sT/2}\right)^m}{2^m s^2\,(Ls^2 + g)}.
\end{aligned}
$$

By taking an inverse Laplace transform for $\Theta(s)$, the real oscillation function $\theta(t)$ can be determined by

$$
\theta(t) = P(t)\beta_0 + Q(t)\beta_1, \tag{16}
$$

where $P(t)$ and $Q(t)$ are inverse Laplace transforms of $\mathcal{P}(s)$ and $\mathcal{Q}(s)$, respectively. Their expressions are specified by (17) and (18) as shown at the top of this page, in which $H(.)$ is the Heaviside function. Having the explicit formula of the oscillation function of the system, we now can integrate (4) to determine the cart's position $x(t)$ and speed $\dot{x}(t)$. The result is summarized in the following Theorem.

Theorem 1. Consider the system (4) controlled by the control law (12) and the nth-order time-polynomial cable

length function $l(t)$ in (14), for all $n \in \mathbb{N}$ and $n \geq 3$, the following results hold

$$\dot{x}(t) = \sum_{i=2}^{n} i(i-1)\alpha_i \int_0^t \sigma^{i-2}\theta(\sigma)d\sigma - g\int_0^t \theta(\sigma)d\sigma - \dot{\kappa}. \tag{19}$$

$$\begin{aligned} x(t) = t\left[-g\int_0^t \theta(\sigma)d\sigma + \sum_{i=2}^{n} i\,(i-1)\,\alpha_i \int_0^t \sigma^{i-2}\theta(\sigma)d\sigma\right] \\ - \sum_{i=2}^{n} i(i-1)\alpha_i \int_0^t \sigma^{i-1}\theta(\sigma)d\sigma \\ + g\int_0^t \sigma\theta(\sigma)d\sigma - \kappa, \end{aligned} \tag{20}$$

where the term κ is given by $\kappa = l\theta$.

Proof. We first prove the result in (19). Differentiating this equation once yields

$$\ddot{x} = -\ddot{l}\theta - 2\dot{l}\dot{\theta} - l\ddot{\theta} - g\theta + \left(\sum_{i=2}^{n} i(i-1)\alpha_i t^{i-2}\right)\theta. \tag{21}$$

Moreover, from (14) the following result holds

$$\ddot{l} = \sum_{i=2}^{n} i(i-1)\alpha_i t^{i-2}.$$

Substituting the above equation into (21) verifies the system dynamics (4), so (19) is a valid solution of the differential equation (4) under the initial condition (3). Since an initial value problem of a linear differential equation only possesses a unique solution, (19) is indeed the only solution of (4).

In order to prove (20), integrating (19) once yields

$$\begin{aligned} x = -\kappa - g\int_0^t \int_0^\lambda \theta(\sigma)d\sigma d\lambda \\ + \sum_{i=2}^{n} i(i-1)\alpha_i \int_0^t \int_0^\lambda \sigma^{i-2}\theta(\sigma)d\sigma d\lambda. \end{aligned} \tag{22}$$

Applying the Cauchy's formula for repeated integral [21, p. 193] on (22), the following results can be obtained

$$\int_0^t \int_0^\lambda \theta(\sigma)d\sigma d\lambda = \int_0^t (t-\sigma)\theta(\sigma)d\sigma$$
$$\int_0^t \int_0^\lambda \sigma^{i-2}\theta(\sigma)d\sigma d\lambda = \int_0^t (t-\sigma)\,\sigma^{i-2}\theta(\sigma)d\sigma.$$

Substituting the above equations into (22) yields (20). This therefore completes the proof. □

Having now the cart's speed and position, we can apply the remaining transport requirement (5) on (19)–(20). As a consequence, the following conditions need to be enforced

$$\begin{aligned} \sum_{i=2}^{n} i(i-1)\alpha_i \int_0^{\tau_f} t^{i-2}\theta(t)dt - g\int_0^{\tau_f} \theta(t)dt = 0 \\ \sum_{i=2}^{n} i(i-1)\alpha_i \int_0^{\tau_f} t^{i-1}\theta(t)dt - g\int_0^{\tau_f} t\theta(t)dt = -x_d. \end{aligned} \tag{23}$$

Since $\theta(t) = P(t)\beta_0 + Q(t)\beta_1$ (see (16)), the equation (23) can be rewritten as

$$\begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 0 \\ -x_d \end{bmatrix}. \tag{24}$$

The terms A_{ij} $(i, j = 1, 2)$ are given as follows

$$\begin{aligned} A_{11} &= \sum_{i=2}^{n} i(i-1)\alpha_i\Gamma_0(i-2) - g\Gamma_0(0) \\ A_{12} &= \sum_{i=2}^{n} i(i-1)\alpha_i\Gamma_1(i-2) - g\Gamma_1(0) \\ A_{21} &= \sum_{i=2}^{n} i(i-1)\alpha_i\Gamma_0(i-1) - g\Gamma_0(1) \\ A_{22} &= \sum_{i=2}^{n} i(i-1)\alpha_i\Gamma_1(i-1) - g\Gamma_1(1), \end{aligned}$$

where Γ_0 and Γ_1 are specified by

$$\Gamma_0(i) = \int_0^{\tau_f} t^i P(t)dt, \;\; \Gamma_1(i) = \int_0^{\tau_f} t^i Q(t)dt.$$

By using the expressions of $P(t)$ and $Q(t)$ in (17) and (18), the integrals Γ_0 and Γ_1 can be computed by the recursive formulas (25) and (26), respectively, as shown at the top of the next page. Since the integrals Γ_0 and Γ_1 are now known, the terms A_{ij} $(i, j = 1, 2)$ in (24) can easily be computed. Therefore, the two unknowns β_0 and β_1 of the unshaped command $u_r(t)$ can be determined to fulfill the remaining transport requirement (5). By this, the control design process is accomplished.

Remark 1. From (25)–(26), it can be seen that the integrals Γ_0 and Γ_1 are polynomials with respect to the unshaped terminal time t_f. Moreover, from (15), the coefficients α_i $(i = 0, \ldots, n)$ are polynomial fractions with respect to t_f. Therefore, the determinant of the matrix $\boldsymbol{A} = [A_{ij}]$ (denoted as $\det(\boldsymbol{A})$) in (24) is also a polynomial fraction of t_f. Consequently, the equation $\det(\boldsymbol{A}) = 0$ has a finite number of isolated roots, as a result of the Fundamental Theorem of Algebra [22, Theorem D.4]. This shows that the system of linear equations (24) has a finite number of isolated values of t_f at which solutions of β_0 and β_1 cannot be found. However, in such a context, we can shift the unshaped terminal time t_f away from the singular points because they are finite in number and isolated.

IV. Simulation results

In this section, two simulations will be conducted to verify the validity of the proposed input shaping-based control law. In all subsequent simulations, the actual nonlinear time-varying crane system (2) is employed for all scenarios.

A. Simulation 1

By this simulation, we wish to show that a non-robust input shaper can be used in the proposed approach to provide a complete vibration suppression at the end of the transportation. Therefore, $m = 1$ is chosen (see (10)) meaning that the

$$
\begin{cases}
\Gamma_0(i) = \dfrac{1}{2^m g}\left[\displaystyle\sum_{k=0}^{m} C_m^k \dfrac{\left(t_f + \dfrac{kT}{2}\right)^{i+1} - \left(\dfrac{kT}{2}\right)^{i+1}}{i+1} - \mu(i)\right] \\
\mu(i) = -\dfrac{i(i-1)}{\omega^2}\mu(i-2) - \dfrac{i}{\omega^2}\displaystyle\sum_{k=0}^{m} C_m^k \left[\left(\dfrac{kT}{2}\right)^{i-1} - \left(t_f + \dfrac{kT}{2}\right)^{i-1}\right] \\
\mu(0) = \mu(1) = 0.
\end{cases} \tag{25}
$$

$$
\begin{cases}
\Gamma_1(i) = \dfrac{1}{2^m g}\left[\displaystyle\sum_{k=0}^{m} C_m^k \left(\dfrac{\left(t_f + \dfrac{kT}{2}\right)^{i+2} - \left(\dfrac{kT}{2}\right)^{i+2}}{i+2} - \dfrac{kT}{2}\dfrac{\left(t_f + \dfrac{kT}{2}\right)^{i+1} - \left(\dfrac{kT}{2}\right)^{i+1}}{i+1}\right) - \phi(i)\right] \\
\phi(i) = -\dfrac{i(i-1)}{\omega^2}\phi(i-2) + \dfrac{1}{\omega^2}\displaystyle\sum_{k=0}^{m} C_m^k \left[\left(\dfrac{kT}{2}\right)^{i} - \left(t_f + \dfrac{kT}{2}\right)^{i} + i t_f \left(t_f + \dfrac{kT}{2}\right)^{i-1}\right] \\
\phi(0) = \phi(1) = 0.
\end{cases} \tag{26}
$$

generalized input shaper is now a standard (non-robust) zero-vibration (ZV) input shaper [2]. The proposed input shaping-based controller utilizes a constant nominal rope length of $L = 0.5\,[l(0) + l(\tau_f)]$, and the terminal time of the unshaped command $u_r(t)$ (see (11)) is chosen as $t_f = 3$ s. A simple hoisting up operation is employed, where $n = 3$ (see (14)), $l_0 = l(0) = 1.5$ m, and $l_1 = l(\tau_f) = 0.5$ m. Therefore, the shaped terminal time is $\tau_f = 5$ s. The reference value for the cart position is $x_d = 2$ m. Using all of the above parameters, essential coefficients for the cable length function $l(t)$ and the unshaped command $u_r(t)$ can be determined by solving (15) and (24). The results are

$$
\begin{aligned}
&\alpha_0 = 1.5,\ \alpha_1 = 0,\ \alpha_2 = -0.1872,\ \alpha_3 = 0.0312, \\
&\beta_0 = 1.3210,\ \beta_1 = -0.8891.
\end{aligned} \tag{27}
$$

The computational time is 0.073 s on a personal computer with 2.7-GHz Intel Core i5-4310M processor and 8 GB of RAM using MATLAB. Therefore, an online implementation is feasible for the proposed controller.

The proposed input shaping controller (12) will be compared with standard input shaping control schemes using ZV/ZVD input shapers [1]. Note that, the (mildly robust) ZVD input shaper stands for zero-vibration-derivative input shaper, and its expression can be found in (10) with $m = 2$. In the case of standard input shaping controllers, the control law $\bar{u}(t)$ in (9) is directly applied to the crane system (2), i.e., they use $u(t) = \bar{u}(t)$. A similar cable length function $l(t)$ with coefficients given in (27) is used for all cases.

It is also worth noting that all schemes (i.e., the proposed input shaping-based control law and the standard input shaping controllers) are compelled to have the same total transfer time. This enforcement is needed to draw a fair comparison between scenarios. As a result, the performance of each approach can be assessed by the amount of residual vibration at the completion of transportation without bias.

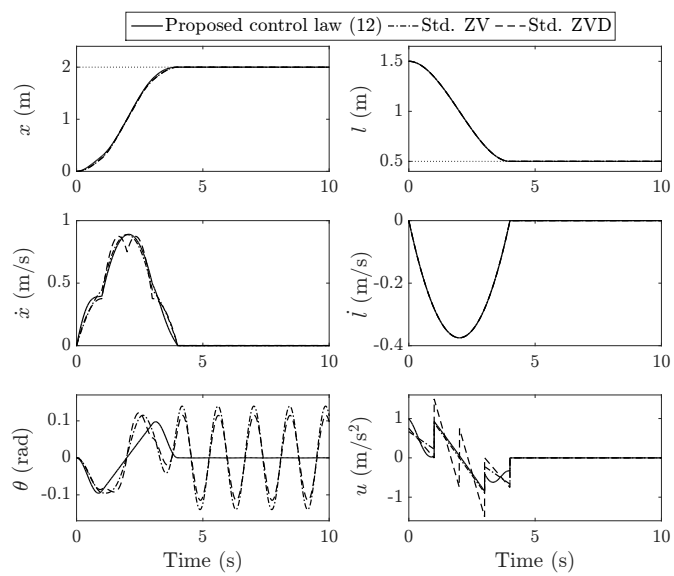

Fig. 3. The comparative simulation result of simulation 1.

The comparative simulation result is depicted in Fig. 3. It can be seen that the proposed input shaping-based controller successfully meets all of transportation requirements specified in (5)–(7). Notably, the proposed controller is able to completely eliminate the residual vibration at the end of the maneuver. In contrast, substantial vibrations persist in the case of standard input shaping control schemes.

B. Simulation 2

The main purpose of this simulation is to verify the correctness of the control design and computation conducted in section III. Hence, a high value of m (see (10)) is chosen, i.e., it is selected that $m = 10$. The unshaped terminal time

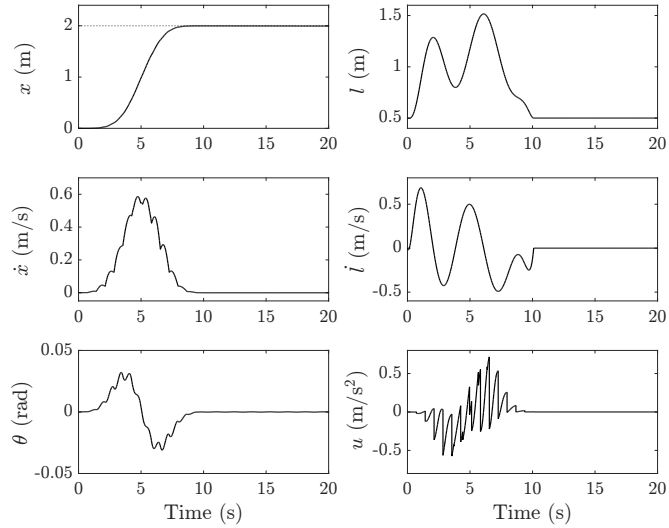

Fig. 4. The result of simulation 2 under a complex hoisting maneuver.

t_f and the reference value for the cart motion are still 3 s and 2 m, respectively. The nominal rope length is $L = 0.5$ m. The crane is put in a working situation requiring a highly complex hoisting maneuver, where the cable length function $l(t)$ must pass through the following nodes

$$(0.5,\ 1,\ 1.2,\ 0.8,\ 1.2,\ 1.5,\ 1,\ 0.7,\ 0.5)$$

at the following time instants

$$\left(0,\ \frac{\tau_f}{8},\ \frac{2\tau_f}{8},\ \frac{3\tau_f}{8},\ \frac{4\tau_f}{8},\ \frac{5\tau_f}{8},\ \frac{6\tau_f}{8},\ \frac{7\tau_f}{8},\ \tau_f\right).$$

By solving (14) and (24), the essential coefficients for $l(t)$ and $u_r(t)$ are given as follows

$$\alpha_0 = 0.5,\ \alpha_1 = 0,\ \alpha_2 = -0.2647,$$
$$\alpha_3 = 1.4639, \alpha_4 = -1.2794,\ \alpha_5 = 0.4955,\ \alpha_6 = -0.1042,$$
$$\alpha_7 = 0.0127, \alpha_8 = -8.92 \times 10^{-4},$$
$$\alpha_9 = 3.34 \times 10^{-5},\ \alpha_{10} = -5.08 \times 10^{-7},$$
$$\beta_0 = -1.3376,\ \beta_1 = 0.884.$$

The computational time is 0.133 s. The simulation result is illustrated in Fig. 4. It can be observed that the proposed controller is able to fulfill all of transportation requirements described in (5)–(7). Therefore, the correctness of control design process performed in section III is firmly confirmed.

V. Conclusion and Future Work

This paper introduces a new input shaping-based controller for overhead crane systems with time-varying cable length. A generalized input shaper and general hoisting motions can be accommodated by the proposed approach. It has been shown that the computational time is small enough for the proposed controller to be implemented in an online fashion. Through simulation results, the proposed scheme demonstrates its superiority to the standard input shaping controllers in the sense that the residual vibration can completely be eliminated at the end of maneuvers. In our future work, we will apply the established input shaping-based control law on an actual crane to verify its practicability. Moreover, unique characteristics of the established technique will be investigated.

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